



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018

**FOURTH YEAR FIRST SEMESTER EXAMINATION FOR
THE DEGREE OF BACHELOR OF SCIENCE AND
BACHELOR OF EDUCATION SCIENCE WITH
INFORMATION TECHNOLOGY**

MAIN CAMPUS

SZL 401: HISTORY AND PHILOSOPHY OF BIOLOGY

Date: 12th March, 2018

Time: 12.00 - 3.00pm

INSTRUCTIONS:

- Answer ANY FIVE Questions
- Illustrate your answers with well labeled diagrams where appropriate



Answer any five (5) questions from this section (20 marks each)

1. Discuss the influence of Christianity on the growth of biology during the middle ages.
2. Give an analysis of the anatomical research of Herophilus (330–250BC) and Erasistratus (310–250 BC).
3. Give an account on the contribution of Galen (130–210AD) to the development of anatomy and experimental physiology.
4.
 - a. Elaborate the statement “Historians and scientists are two groups of scholars with different backgrounds and viewpoints yet both have considered the history of biology as their own”. 10marks
 - b. Explain the problem solving approach to studying the history of biology. 10 marks
5. Discuss the biological works of Aristotle (384–322BC).
6. Discuss the evidence of biological science and medicine in ancient Mesopotamia.
7. Discuss the extent to which the following statement is factual: “The major historical contribution of Islamic medicine was preservation of ancient Greek wisdom”.
8. Explain the significance of the following in scientific method;
 - a. Observation
 - b. Asking The Right Question
 - c. Experiment.



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MAIN CAMPUS

SZL 403: PARASITOLOGY I

Date: 13th February, 2018

Time: 8.30 - 11.30 am

INSTRUCTIONS:

- Answer ALL questions in SECTION A and ANY THREE questions from SECTION B.



SECTION A [40 Marks]

Answer ALL questions in this Section. Each question carries 5 marks.

Q1. State any five (5) ways that depict how to prevent toxoplasmosis due to contamination with cat faeces.

Q2. State any five (5) ways by which protozoan parasites may be transmitted to a new host. In each case, give one (1) example of the parasite involved.

Q3. Name any five (5) factors that contribute to increased risks of *Cryptosporidium* water-borne outbreaks.

Q4. *Trichomonas vaginalis* is the most common pathogenic protozoan infection of humans in industrialised countries. State the following:

- a) The form of the parasite responsible for the transmission of the disease.
- b) **Two (2)** anatomical locations, each, for the female and male, where the parasite replicates.
- c) The most practical and rapid method of diagnosis; what it detects.
- d) The most sensitive method of diagnosis; its drawback.
- e) **Two (2)** categories of persons among whom there is higher prevalence of trichomoniasis.
- f) The **prominent symptom** in women with *Trichomonas vaginalis* infection.

Q5. Parasitic protozoa are classified into various phyla. Name any five (5) of the phyla; in each Phylum, state one (1) member.

Q6. Name five (5) important vectors of *Trypanosoma cruzi* in South America. For each vector, highlight foci of transmission of Chagas' disease.

Q7. A parasite is metabolically or physiologically dependent on its host. State four (4) requirements for which a parasite may be dependent on its host. For each requirement, give one (1) example of the relevant parasite.

Q8. State any five (5) reasons why a human foetus is NOT a parasite.

SECTION B [60 marks]

Answer **ANY THREE (3)** questions in this Section. Each question carries 20 marks.

Q9. Give a detailed account of the pathogenesis of *Plasmodium falciparum* infection in human, from the time a female *Anopheles* mosquito bites and deposits sporozoites in its human host to the time disease, malaria, manifests; highlight key stages in the development.

Q10. Discuss the pathogenesis of the mild-to-profuse watery diarrhoea that is the most common clinical manifestation of cryptosporidiosis.

Q11. Describe the life cycle of *Sarcocystis cruzi* in its various hosts, including humans.

Q12. Discuss strategies for the control of parasitic diseases.

Q13. Discuss African sleeping sickness and South American trypanosomiasis, emphasising differences where applicable.



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INFORMATION TECHNOLOGY**

MAIN CAMPUS

SZL 405: AGRICULTURAL ENTOMOLOGY

Date: 14th February, 2018

Time: 8.30 - 11.30am

INSTRUCTIONS:

- Answer ALL Questions in Section A and any other THREE in Section B
- Use illustrations where appropriate



SECTION A (40 MARKS)

ANSWER ALL QUESTIONS FROM THIS SECTION (5 MARKS EACH)

1. Distinguish between two modes of insect development and explain two advantages of each mode giving examples.
2. Insect are known to be economically disturbing. Discuss how true this is giving specific examples.
3. Insects evolved from the same ancestor with other arthropods. Prove this giving specific examples.

4. Explain why invasive pest species become very destructive causing heavy losses of crops.
5. Define and explain how economic threshold relates to pest management in an agroecosystem.
6. Explain clearly three (3) mechanisms of resistance re exhibited in host plant resistance strategy of pest control.
7. Biological control uses three (3) agents in any of the three named strategies. Explain how these are done.
8. Explain how three named insecticide additives enhance the use and effectiveness of insecticides giving examples

SECTION B (60 MARKS)

Answer any THREE questions from this section (20 Marks each)

9. Discuss in detail the reasons why in spite of being very effective in pest control insecticides are being progressively rejected.
10. Describe how you would apply the three strategies of biological control using each of the three agents giving examples.
11. Discuss the advantages of Genetically Modified Organisms (GMO) and explain the reasons for rejection by most people.
12. Discuss anthropogenic activities have affected insect pollinators and food security in the twenty first century
13. Describe the role of Insect exoskeleton in the defense of most pests of agricultural importance



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MAIN CAMPUS

SZL 409: FISH AND FISHERIES BIOLOGY

Date: 21st February, 2018

Time: 8.30 - 11.30 am

INSTRUCTIONS:

- Answer ALL questions in SECTION A and any THREE from SECTION B



SZL 409: FISH AND FISHERIES BIOLOGY

INSTRUCTIONS:

1. Answer ALL questions in section A (5 Marks each)
 2. Answer any THREE questions in section B (20 Marks each)
 3. Use illustrations where appropriate
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SECTION A (40 Marks):

Q1. Describe the adaptations of marine fishes to extended stay under water during diving

Q2. State the FIVE main distinctive characteristics of the Superclass Agnatha

Q3. Briefly describe Five (5) morphological characteristics used in fish identification

Q4. Explain the term 'population dynamics' and describe the three (3) dynamic rate functions

Q5. State FIVE characteristics that distinguish the Class Chondrychthyes from the Class Osteichthyes

Q6. Describe Stokes formula for terminal sinking velocity and explain its application in floatation in fishes

Q7. Explain how fishes use echolocation for survival in their environment

Q8. Using a graphic representation, describe the classification of modern day fishes

SECTION B:

INSTRUCTIONS: Answer any THREE questions (20 Marks each question)

Q9. Describe the circulatory system and gaseous exchange in Teleostean fishes

Q10. Describe the main trophic categories and feeding adaptations in fishes

Q11. Citing specific examples, describe the principal fish migration patterns

Q12. Describe the structural and functional adaptations of fishes for locomotion in water

Q13. Describe the different modes of reproduction and reproductive strategies of fishes



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MAIN CAMPUS

SZL 415: LIMNOLOGY

Date: 20th February, 2018

Time: 8.30 - 11.30 am

INSTRUCTIONS:

- Answer ALL questions in SECTION A and any TWO from SECTION B



SZL 415: LIMNOLOGY

Date:

Time: 3 Hrs

INSTRUCTIONS:

1. Answer ALL questions in section A (4 marks each).
2. Answer any TWO questions in Section B (15marks each).
3. Use illustrations where possible.

SECTION A: ANSWER ALL QUESTIONS IN THIS SECTION (4 MARKS EACH)

1. State the bases on which all aquatic organisms are classified, mentioning the classes with an example each.
2. Distinguish between: a) oligotrophic and dystrophic lakes and b) paleolimnology and lake history.
3. State the key steps in tackling sedimentation crisis in a given water body.
4. Briefly describe any four possible water circulation patterns in lakes.
5. State the factors that influence the chemical and biological characteristics of a lake.
6. Briefly describe the term 'nutrients' in a water body.
7. State the key organism structure of a freshwater ecosystem.
8. Explain the term 'acidification' in reference to freshwater environments.
9. State the key factors responsible for thermal stratification in temperate lakes.
10. Briefly describe any four physico chemical properties of water and their determination.

SECTION B: ANSWER ANY TWO QUESTIONS (15MARKS EACH).

11. Discuss the watershed operations that adversely impact inland aquatic resources.
12. Describe the various methods/processes by which lakes are formed.
13. Outline any chemical method by which oxygen can be determined in a water body.
14. Discuss the possible key considerations in the ecology and management of a named water body.